

Lighting Bar Grid

Technical Datasheet - Theatre Compliance Portal

Document type: Equipment technical information / inspection reference

Equipment name	Lighting Bar Grid
Equipment category	Lifting Equipment / Structural Support
Portal class ID	structural_support
Typical application	Fixed or suspended lighting bar grid used to support luminaires, cabling and associated production equipment.
Primary considerations	Structural integrity, load path, support points, fixings, SWL/WLL, cable management, safe access and inspection records.
Relevant framework	LOLER, PUWER, BS EN 17206 and ABTT Technical Standards / good practice where applicable.
Datasheet status	Generic template - site-specific loads, dimensions and design information must be confirmed by the competent person or designer.

1. Description

A lighting bar grid is a fixed or suspended arrangement of lighting bars used within a theatre, studio or entertainment venue to support luminaires, cable runs, accessories and other approved production equipment. The grid, its suspension points, fixings, brackets and supporting structure should be considered as one complete load path.

2. Intended Use

- To support luminaires and approved production equipment within the stated design limits.
- To provide a predictable and managed arrangement of lighting bars and load attachment points.
- To allow safe installation, maintenance and inspection of lighting equipment by competent persons.
- To support associated cable management where the grid design allows it.

3. Technical Data to Confirm

Grid location / venue reference	To be completed
Bar diameter / section	To be confirmed
Material / finish	To be confirmed
Overall dimensions	To be confirmed
Support method	To be confirmed
Number and type of suspension points	To be confirmed
Safe Working Load / Working Load Limit	To be confirmed from design documentation
Point-load limits	To be confirmed
Distributed-load limits	To be confirmed
Electrical containment / internally wired sections	To be confirmed
Inspection interval / scheme reference	To be confirmed
Competent person / responsible person	To be completed

4. Operational Limitations

- Do not exceed the stated Safe Working Load / Working Load Limit.
- Do not apply concentrated point loads unless the grid has been designed and verified for that loading.
- Do not drill, weld, modify or add suspension points without approval from a competent person or designer.
- Loads should be evenly distributed where possible.
- All clamps, safety bonds and accessories must be suitable for the load and compatible with the bar or grid section.
- Any change of use, additional loading, damage, impact or modification should trigger review before further use.

5. Inspection Checklist

Check item	Inspection guidance	Status / notes
General condition	Check bars for bending, dents, cracks, corrosion, deformation, excessive wear or signs of overloading.	
Fixings and supports	Check visible bolts, clamps, brackets, hangers, drop rods, chains, beam clamps or other supports are present, secure and correctly fitted.	
Load path	Confirm the grid, suspension points, fixings and supporting structure are treated as one complete load path.	
SWL / WLL information	Confirm load information is known, legible, suitable and reflected in local documentation.	
Connections and joints	Check welded joints, bolted connections and tube intersections for cracks, movement, missing fixings or loose fittings.	
Suspended equipment	Confirm luminaires and other equipment are attached using suitable clamps and secondary suspensions where required.	
Cable management	Check cables are supported and not causing snagging, abrasion, strain, trip or access risks.	
Modification evidence	Look for unauthorised drilled holes, added fixings, non-standard repairs, damaged coatings or corrosion.	
Records	Check that inspection records, maintenance history and defect actions are current and available.	

6. Maintenance and Defect Action

- Remove affected areas from service if serious damage, deformation, missing fixings or uncertain load capacity is identified.
- Record all defects and escalate to a competent person for review.
- Do not return the grid to service until defects have been assessed and remedied.
- Maintain the grid in accordance with manufacturer, designer or competent-person instructions.
- Review inspection frequency where use is heavy, conditions are harsh, or defects are repeatedly identified.

7. Documentation to Retain

- Design information, drawings or load calculations where available.
- SWL / WLL information, point-load limits and distributed-load limits.
- Installation records, commissioning records and modification records.
- Inspection and thorough examination reports where applicable.
- Maintenance records and defect close-out evidence.
- Manufacturer instructions or competent-person guidance.

8. Portal Summary Text

Suggested short description for the portal:

A lighting bar grid is a fixed or suspended arrangement of lighting bars used to support luminaires, cabling and associated production equipment. The grid, its support points and fixings should be assessed as one complete load path and used only within confirmed load limits.

9. Disclaimer

Important: This datasheet is a generic technical information template. It does not replace manufacturer instructions, design calculations, site-specific risk assessment, formal inspection records, or advice from a competent person. Site-specific dimensions, load ratings and limitations must be verified before use.